

Volker Scheidemann

Introduction to Complex Analysis in Several Variables

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Preface

The idea for this book came when I was an assistant at the Department of Mathematics and Computer Science at the Philipps-University Marburg, Germany. Several times I faced the task of supporting lectures and seminars on complex analysis of several variables and found out that there are very few books on the subject, compared to the vast amount of literature on function theory of one variable, let alone on real variables or basic algebra. Even fewer books, to my understanding, were written primarily with the student in mind. So it was quite hard to find supporting examples and exercises that helped the student to become familiar with the fascinating theory of several complex variables.

Of course, there are notable exceptions, like the books of R.M. Range [9] or B. and L. Kaup [6], however, even those excellent books have a drawback: they are quite thick and thus quite expensive for a student's budget. So an additional motivation to write this book was to give a comprehensive introduction to the theory of several complex variables, illustrate it with as many examples as I could find and help the student to get deeper insight by giving lots of exercises, reaching from almost trivial to rather challenging.

There are not many illustrations in this book, in fact, there is exactly one, because in the theory of several complex variables I find most of them either trivial or misleading. The readers are of course free to have a different opinion on these matters.

Exercises are spread throughout the text and their results will often be referred to, so it is highly recommended to work through them.

Above all, I wanted to keep the book short and affordable, recognizing that this results in certain restrictions in the choice of contents. Critics may say that I left out important topics like pseudoconvexity, complex spaces, analytic sheaves or methods of cohomology theory. All of this is true, but inclusion of all that would have resulted in another frighteningly thick book. So I chose topics that assume only a minimum of prerequisites, i.e., holomorphic functions of one complex variable, calculus of several real variables and basic algebra (vector spaces, groups, rings etc.). Everything else is developed from scratch. I also tried to point out some of the relations of complex analysis with other parts of mathematics. For example, the Convergence Theorem of Weierstrass, that a compactly convergent sequence of holomorphic functions has a holomorphic limit is formulated in the language of

functional analysis: the algebra of holomorphic functions is a closed subalgebra of the algebra of continuous functions in the compact-open topology.

Also the exercises do not restrict themselves only to topics of complex analysis of several variables in order to show the student that learning the theory of several complex variables is not working in an isolated ivory tower. Putting the knowledge of different fields of mathematics together, I think, is one of the major joys of the subject. Enjoy !

I would like to thank Dr. Thomas Hempfling of Birkhäuser Publishing for his friendly cooperation and his encouragement. Also, my thanks go to my wife Claudia for her love and constant support. This book is for you!